	Engineering Work Order for In-kind Replacement (Non MOC) **Инженерная Инструкция для Работ по Аналогичной Замене (Без КЗИ)**							
	Anomaly Report: Отчет об аномалии:	N/A	**EWR #:** # ЗИР:	25-0114	**Subsystem:** Подсистема:	TL	**Line №:** № Линии:	62-0400-TL-2148-1 1/2"-150H22-HCW5
	Area: Участок:	SGP	**WO:** РЗ:	1665591	**P&ID #:** СТиКИП	62-0400-B-PID-0429-04	**Title:** Название:	TA26 TLR#22263 REMOVAL ON TL LINE
TENGIZCHEVROIL ТЕНГИЗШЕВРОЙЛ	**Unit:** Установка:	U400	**Drawing/Sketch No:** Чертеж/Схема No:	062-0400-LLL-ISO-20083-01				
Standard of Acceptance: Стандарт Приемки:	B31.3		**TCO Specification:** Спецификации ТШО:	W-ST-2025			**TCO Approved WPS:** Утвержденная Сварочная ПроцедураТШО:	20AW
Is wall thickness measurement required at tie-in points prior to start works (to verify existing pipe sound wall thickness) Требуется ли замер толщины стенки на точках врезки перед началом работ (для определения приемлемой толщины существующего трубопровода)			YES	**Is 100% PT/MT required for weld bevel ends prior to welding** Требуется ли 100% ККД/МПД для проверки кромок перед сваркой				NA/НП
Shop Hydro-test Испытание Гидро-тестом в Цеху	NA/НП barg		**Piping NDT** НК Трубопровода	NA/НП VT/ВК 100%	**Coating System #:** № Системы Покрытия:	12.1	**Hydrogen Bake out** Водородный Отжиг	NA/НП
Field Hydro-test Испытание Гидро-тестом на Участке	NA/НП barg		**Piping Support Structural NDT** НК Опорной Конструкции Трубопровода	NA/НП	**Insulation type and thickness:** Тип и толщина изоляции:	HCW 5 - 30mm	**PMI for Alloy materials (SP 20)** ХАМ для Материалов из сплавов (ПТБ 20)	NA/НП
HARDNESS TESTING (Maximum up to 200HB) ОПРЕДЕЛЕНИЕ ТВЕРДОСТИ (Максимум до 200HB)	NA/НП		**Stress Relief (PWHT)** Снятие Напряжений (ПСТО)	NA/НП	**Notes** Примечания	SERVICE TEST		

Job Description
Описание работ

Please, see next pages for step associated with work execution/ Прошу ссылаться на следующие страницы по выполнению работ.

Approved by Потверждено		**Name** Имя	**Signature** Подпись
Originated by: Подготовлено:	**DE MPS Engineer** Инженер DE MPS	Adil Maxutov	Adil Maxutov Digitally signed by Adil Maxutov Date: 2025.03.02 09:13:15 +05'00'
Approved by: Утверждено:	**DE Engineer-Consultant** DE Инженер-Консультант проектной группы		ucrx, Rakhymzhan Saginayev Digitally signed by ucrx, Rakhymzhan Saginayev Date: 2025.03.07 08:32:37 +05'00'
Reviewed by: Рассмотрено:	**RE QM Lead:** Вед. Специалист КК Отдела Надежности:		
Reviewed by: Рассмотрено:	**FER Authorized Inspector:** Уполномоченный инспектор НСО:		
Reviewed by: Рассмотрено:	**TA Group Supervisor:** Руководитель группы КР:		
Reviewed by: Рассмотрено:	**TA Unit Coordinator:** Координатор КР установок ОЭ:		
Approved by: Утверждено:	**Design Engineering Supervisor:** Руководитель инженерно-проектной группы:		

PRESENT SITUATION:

- Replace damages section of TL condensate line 62-0400-TL-2148-1 1/2"-150H22-HCW5 on U400. TLR# 22263, WO# 1651305.

PROPOSED SOLUTION:

- Replace damaged TEE of TL condensate line 62-0400-TL-2148-1 1/2"-150H22-HCW5.

PRELIMINARY SITE WORK:

- Prior to the start of the mechanical activities, perform wall thickness testing at Tie-in Points (TP) showed in isometric drawing 062-0400-LLL-ISO-20083-01 to confirm that wall thickness is adequate for welding.

  > **NOTE:** Results shall be reviewed and approved by the Fixed Equipment Reliability (FER) Unit Inspector.

- Prior to commencing any fabrication works verify dimensions on site.

SHOP WORKS:

- Withdraw materials required for this EWO from TCO Warehouse according to Material Request(s).
- Abrasive blast and coat new piping, as specified on isometric drawing 062-0400-LLL-ISO-20083-01 in accordance with TES COM-SU-5191-TCO.

SITE WORKS:

- Perform relevant job safety assessments (PPHA & JSA) and obtain required Permits-to-Work (PTWs).
- Demarcate work area as necessary.

  > **NOTE**: TCO Operations shall isolate, depressurize, drain and ready for Hot Work, Equipment and Piping associated with this Project, if and as applicable, and shall hand it over to CONTRACTOR; fully accessible, safe and Locked-Out-Tagged-Out (LOTO) as scheduled and agreed between TCO Operations and CONTRACTOR.

- Remove all cladding and insulation to the extent required.
- Zone Maintenance to isolate, LOTO and remove electric trace heating and instruments from associated piping
- Cut and remove existing piping shown on "Destruct Detail" of isometric drawing 062-0400-LLL-ISO-20083-01.
- Bevel cut ends of existing piping at the tie-in points for field welding as shown on isometric drawing 062-0400-LLL-ISO-20083-01.
- Inspect the prepared weld bevel ends 100% VT and 100 % MT/PT for defects.

  > **NOTE**: Perform inspection of weld bevels PRIOR to welding.

- Install piping as specified on isometric drawing 062-0400-LLL-ISO-20083-01. For field welds use TCO-approved WPS indicated in on isometric drawing.
- Perform NDE of field welds as specified in on isometric drawing 062-0400-LLL-ISO-20083-01.
- Perform surface preparation and touch up field welds and any other coating damages in accordance with TES COM-SU-4743-TCO and coating system(s) per TES COM-SU-5191-TCO specified on drawings.
- Arrange with Zone Maintenance to reinstate related electric trace heating.
- All QM piping passport documentation (A-category Checklists) to be verified and signed by TCO-approved QM Inspector, but not later than 24 hours after mechanical scope completion.
- Advise Mursal Rauan / Ramazanov Nurbulat DE TA26 SGP/SGI Mechanical Engineers at ext. 4259 about work completion and invite all necessary people to PSSR.
- Job Pack Coordinator shall submit the completed PIC checklist to Operations department after being signed by Quality Management group. (Form is provided by QM Dept.).
- Conduct PSSR and close out A-category punch points.
- Reinstate insulation and cladding as per IRM-SU-1381 as specified in EWO.
- Close out all B-category punch points
- Clean up the work area and demobilize from site.

AFTER COMPLETION OF WORK:

- Perform service test.
- Final as-built package (including B-category Checklist) shall be submitted to TCO QM group within 2 weeks after completion of the whole scope of works.

REFERENCED TCO SPECS:

- A-ST-2008 — Basic Engineering Data
- SI-105 — Permit to work and hazard analysis
- W-ST-2025 — Welding, PWHT and NDE of piping
- PIM-SU-2505-TCO — Carbon Steel Piping Fabrication
- RM-SU-1381-TCO — Thermal Insulation for Hot Lines